

Computing before computers

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Most people think of a computer as an electronic device with a screen and a keyboard. However, humans have been using calculating devices for thousands of years.

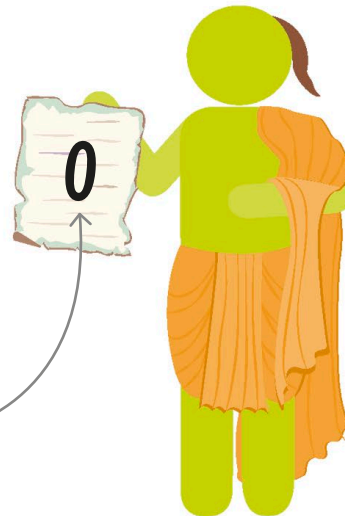
The earliest computers

The earliest calculations were done using the 10 fingers of the hands, which is why most number systems are based on multiples of 10. The Romans had a method for solving complex calculations using their fingers. This practice gave us the word *digit*, from the Latin word *digitus*, meaning finger. It is now used to denote any number between 0 and 9.

▷ Counting numbers

Computers use only the digits 0 and 1 to represent numbers using the binary system, which is based on multiples of two. The concept of zero, as we know it today, originated in India in the 5th century.

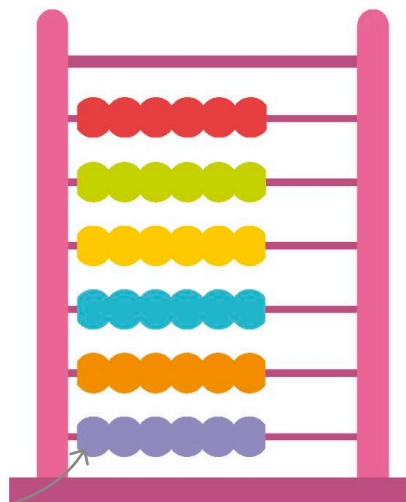
The ancient Indian Bakhshali manuscript contains the first known mention of zero.



Stone structures and counting boards

Early calculating devices were used for determining time. They included ancient Egyptian structures, called obelisks, where large columns of stone cast shadows that moved with the Sun. The direction of the shadows helped people roughly determine the time. The earliest counting boards, which used columns of metal or stone discs (but had no wires), appeared in about 2400 BCE in Babylon (modern-day Iraq). Here are two other examples of early calculating aids.

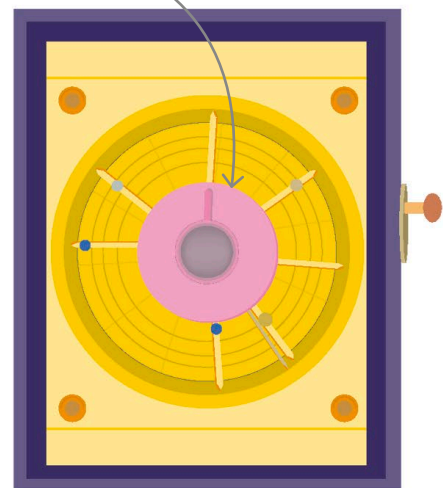
Movable beads strung on wires



△ Abacus

Building on the invention of the counting board, the abacus used beads on wires to represent units, tens, hundreds, and so on.

Bronze dial showing constellations of the zodiac



△ Antikythera mechanism

Found off the coast of the Greek island of Antikythera, this ancient Greek clock-like mechanism has 37 gears, and was used to calculate the positions of stars and planets, and predict eclipses.